POLICY COMPLIANCE INTELLIGENCE SYSTEM ARCHITECTURE

End-to-End Data Processing: Policy Ingestion -> Hybrid Retrieval -> Multi-Agent Workflow -> Compliance Dossier

1. Policy Data & Knowledge Base

Task 1: Corpus Prep & Evidence Storage

- Enterprise PDF Policy Corpus (7 Policies)
- Document Extractor & Section Parser
- Metadata Enrichment (Frameworks, Severity)
- Dense Embeddings (all-MiniLM-L6-v2)
- Sparse BM25 / Sublinear Lexical Index
- SQLite Metadata & Relational Store
- Permanent Vector Cache (embeddings_cache.npz)
- Verbatim Regulatory Clause Repository

2. Hybrid Retrieval Engine

Task 2: Evidence Search & Citations

- Natural Language Compliance Search API
- Dense Vector Cosine Similarity Matcher
- Sparse BM25 Exact Keyword Matcher
- Reciprocal Rank Fusion (RRF Re-Ranking)
- Domain & Category Metadata Filtering
- Similarity Threshold Gatekeeper (>0.28)
- Evidence Citation Engine [POL §X.X]
- Verbatim Snippet & Key-Term Highlighting

3. Multi-Agent Analysis Workflow

Task 3: Collaborative Agent System

- Policy Retrieval Agent (Intent & Discovery)
- Compliance Analysis Agent (Clause Evaluation)
- Risk Assessment Agent (Penalty Matrix)
- Policy Validation Agent (Faithfulness Audit)
- Recommendation Agent (Remediation Plan)
- A2A Communication Bus (Structured Messages)
- Dynamic Escalation & Specialist Handoffs
- State Machine & Execution Trace Logger

4. Compliance Dossier & UI

Task 4 & 5: Microservice & Delivery

- Fast, Responsive Single-Page Web App
- Compliance Verdict (Compliant/Non-Compliant)
- 0-100 Risk Exposure Gauge & Heatmap
- Interactive Clause Assessment Matrix
- Remediation Sign-Off Checklist
- Official PDF Audit Dossier Generator
- Automated Benchmark Suite (10 Scenarios)
- REST Endpoints & SSE Streaming API

VERIFIED SYSTEM BENCHMARKS & PERFORMANCE METRICS

• Retrieval MRR: 100.0% • Retrieval Recall@5: 100.0% • Verdict Accuracy: 100.0% • Risk Calibration: 100.0% • Grounding Faithfulness: 98.0% • Latency: ~92 ms

Regulatory Coverage: FinCEN BSA | OFAC | GDPR (Articles 6, 28, 33) | ISO 27001 | SOC 2 | EU AI Act | SEC Rule 10b-5 | FCPA | HIPAA Safe Harbor